FITNESS APP SRS DOCUMENT

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▼ 1.INTRODUCTION

This document provides a comprehensive overview of the functional and non-functional requirements for the Fitness Tracking App. The app is designed to assist users in monitoring their health, fitness, nutrition, and weight loss through personalized workout plans, calorie tracking, and real-time analytics.

▼ 1.1 PURPOSE

The purpose of this document is to define the functional and non-functional requirements of a **fitness app.** This app is designed to help user monitor and track progress towards their health , nutrition ,fitness and weight loss through personalized workout plans, calorie counts and real time analytics.

▼ 1.2 SCOPE

This app will be a mobile application available on android and iOS .

It will allow user following:

1. Customizable workout plans based on their goals
2. Calorie count and step tracking
3. Integration with wearable fitness devices
4. Social feature like progress sharing and weekly challenges
5. Real-time analytics and progress tracking
6. Calendar integration for workout scheduling
7. Push notifications for reminders and goal tracking

▼ 1.3 INTENDED AUDIENCE AND READING SUGGESTIONS

This document is intended for stake holders, software developers , software users and designers, testers , project managers and documentation writers to serve as guide to ensure that the final product meets the user requirements ,specifications and business goals.

▼ 1.4 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

- **SRS** – Software Requirement Specification
- **UI** – User Interface
- **API** – Application Programming Interface
- **OS** – Operating System
- **BLE** – Bluetooth Low Energy

▼ 1.5 REFERENCES

- IEEE Std 830-1998 - Software Requirements Specification Standard
- Google Android and Apple iOS Developer Guidelines
- Health and fitness app regulations (GDPR, HIPAA, etc.)

▼ 1.6 DOCUMENT OVERVIEW

This document is divided into the following sections:

1. **Introduction:** Overview of the document and app purpose.
2. **Product Overview:** Detailed description of the app's functionality and constraints.
3. **Requirement Specification:** Technical details and requirements of the system.
4. Use Cases:

▼ 2.PRODUCT OVERVIEW

this section will provide a in-depth analysis of fitness tracking app, functionalities , constraints , assumptions and more. It serves as a roadmap for development and understanding of app

▼ 2.1 PRODUCT PRESPECTIVE

The Fitness Tracking App is a mobile application available for both Android and iOS. It serves as a personal health and fitness companion, offering users the ability to:

- Track and customize their workout plans
- Monitor calorie intake and steps
- Integrate with wearable fitness devices
- Engage in social fitness challenges
- Receive real-time analytics and progress tracking
- Sync with calendar applications for scheduling

▼ 2.2 PRODUCT FUNCTIONS

The app will include the following core functions:

1. **Customizable Workout Plans:** Users can input their fitness goals and receive tailored workout recommendations.
2. **Integration with Wearable Devices:** Syncing with smartwatches and fitness trackers to collect health data.
3. **Calorie Count and Step Tracking:** Users can log daily calorie intake and track steps automatically.
4. **Calendar Integration:** Users can schedule workouts and receive reminders.
5. **Social Features:** Users can join communities, participate in challenges, and share progress.
6. **Tracking Analytics:** Detailed progress reports, including graphs and insights.
7. **Push Notifications:** Timely reminders for workouts, calorie logging, and fitness milestones.

▼ 2.3 PRODUCT LIMITATIONS

- **Platform Constraints:** Some features may vary between Android and iOS due to OS limitations.
- **Device Compatibility:** The app may have performance variations based on device specifications.
- **Battery Consumption:** Background syncing with wearables may impact battery life.
- **Internet Dependency:** Some features require an active internet connection for cloud sync and social interactions.

▼ 2.4 USER CHARACTERISTICS

- **Beginners:** Users looking for guided workout plans and progress tracking.
- **Intermediate & Advanced Users:** Users who want advanced customization and analytics.

- **Wearable Users:** Users integrating data from smartwatches and fitness bands.
- **Community-Oriented Users:** Users interested in social engagement, challenges, and competition.

3.REQUIRMENT SPECIFICATION

This section details the external interfaces, functional requirements, and non-functional requirements of the system.

3.1 EXTERNAL INTERFACES

3.1.1USER INTERFACE

The user will be intuitive and simple to navigate for both android and iOS platforms. this will include

- **Home Screen** :Displays personalized workout plan, calorie burned , step count and a quick overview of goals and progress.
- **Workout Screen:** shows detailed workout routine with exercise description and sets/reps along with ability to track progress in real time
- **Progress Screen:** Displays analytics ,graphs reports and shows progress over time for calories burned and weight changes ,goals completed
- **Calendar Integration:** Displays workout schedule with calendar synchronization, showing upcoming workouts and rest days.
- **Social Feed:** Community interactions, challenges, and progress sharing.
- **Settings Screen:** Allows user to customize it to their preference details like goals weight etc. , and manage notifications ,reminders, app permission.

3.1.2.HARDWARE INTERFACES

- **Smartphones:** Compatible with Android (Version X and above) and iOS 10+.
- **Wearable Devices:** Supports fitness trackers via Bluetooth or Wi-Fi.

3.1.3 SOFTWARE INTERFACES

- **Mobile OS Compatibility:** Android and iOS platforms.
- **Third-Party APIs:** Google Fit, Apple Health, and wearable device APIs.
- **Cloud Sync:** Secure data storage and synchronization.
- **Calendar Integration:** Sync with Google Calendar and iCal.

▼ 3.2 FUNCTIONAL REQUIREMENTS

▼ 3.2.1 CUSTOMIZABLE WORKOUT PLAN

DESCRIPTION

The app will allow users to input their fitness goals (e.g. weight loss, muscle gain, endurance) and generate a personalized workout plan tailored to their fitness level and preference .These plans will include the type of exercises, number of set, reps and recommended rest intervals.

FUNCTIONAL RESPONSE

The app will offer a series of customizable options to help user specify there workout goals and will automatically adjust the workout plans to fit their specified requirements .

▼ 3.2.2 CALORIE AND STEP TRACKING

DESCRIPTION

The app will enable users to track their daily calorie intake (manually or via integration with nutrition tracking) and count steps throughout the day (using the phone’s built in sensor or integrating with wearable devices)

FUNCTIONAL RESPONSE

The app will allow users to log calories consumed and steps taken manually while also offering automatic syncing of calories and steps from connected devices (e.g. Fitbit , apple Watch)

▼ 3.2.3 INTEGRATION WITH WEARABLE DEVICES

DESCRIPTION

The app will integrate with wearable fitness devices such as **Fitbit , Apple Watch and Garmin etc.** to capture real time data like heart rate, calories burned, steps, and sleep patterns. This data will be used to provide more accurate insights into the users fitness performance

FUNCTIONAL RESPONSE

The app will automatically sync data from connected wearables . User will be able to link their fitness devices to app during setup and sync data will seamlessly on a regular basis .

▼ 3.2.4 CALENDER INTEGRATION

DESCRIPTION

The app will integrate with google calendar and ical to help users schedule workouts , rest days and other fitness related events .The app will send reminders and notifications to users calendar to help them stay on track with their fitness goals

FUNCTIONAL RESPONSE

The app will allow users to set workout schedules , and these will be synced with users calendar providing visible reminders and notifications about upcoming workouts to help them stay on track with their fitness goals .

▼ 3.2.5 SOCIAL FEATURES AND COMMUNITY ENGAGEMENT

DESCRIPTION

The app will provide users with the ability to share progress updates, join challenges, and interact with other users within communities . This will include features like sharing workout achievements, daily/weekly fitness goals, and engaging in challenges or group activities in communities

FUNCTIONAL RESPONSE

The app will have a social feed where users can post updates, challenge others, and track their progress in real-time. Users will be able to comment, like, and share posts within the app.

3.2.6 PUSH NOTIFICATIONS

DESCRIPTION

This app will send notifications to remind user of task such as goals set, workout plans and challenges. Notifications will essentially be customizable by the user example calorie log reminders etc.

FUNCTIONAL RESPONSE

The app will allow user to set preference in notifications and send reminders at particular time for activities like work out, tracking of nutrition, fitness challenges, goals.

3.2.7 PROGRESS REPORTS AND ANALYTICS

DESCRIPTION

This app will provide user with real time progress reports displaying visual graphs &analytics for activities carried out by the user , goals completed , challenges progress , calories burned, steps taken, weight loss. User can view their progress of day, weeks & months.

FUNCTIONAL RESPONSE

The app will provide graph that will show detailed insight into users fitness journey such as goals completed , challenges progress , calories burned ,steps taken ,weight loss etc.

3.3 NON FUNCTIONAL REQUIREMENTS

- **Performance:** The app should load within 3 seconds and handle 100,000 concurrent users. It should support real-time tracking updates with minimal lag.
- **Security:** User data should be encrypted, comply with GDPR/HIPAA, implement multi-factor authentication, and ensure secure login/logout

mechanisms.

- **Scalability:** The backend should support up to 10 million users, allow auto-scaling, and provide efficient data handling for large user traffic.
- **Usability:** The UI should follow best UX design practices, support voice commands, and be accessible for users with disabilities.
- **Reliability:** The app should have a 99.9% uptime, provide offline mode support, and include auto-recovery mechanisms in case of failure.
- **Maintainability:** The system should allow modular component updates, ensure backward compatibility, and provide automated software updates.

▼ 4. USE CASES

▼ 4.1 CUSTOM WORKOUT PLAN CREATION

- **Actors:** User, System
- **Description:** Users can enter their fitness goals, and the system will generate a workout plan.
- **Preconditions:** User account must be created.
- **Steps:**
    1. User inputs fitness goals.
    2. System suggests a workout plan.
    3. User customizes the plan.
    4. Plan is saved.
- **Postconditions:** The workout plan is stored in the user's profile.
- **Exceptions:** If the user does not provide enough data, the system prompts for additional information.

▼ 4.2 STEP TRACKING

- **Actors:** User, System, Wearable Device
- **Description:** The app tracks steps via phone sensors or wearables.

- **Preconditions:** Step tracking must be enabled.
- **Steps:**
    1. User enables step tracking.
    2. The system syncs step data from sensors.
    3. The progress is displayed in the app.
- **Postconditions:** Step count is updated in the progress report.
- **Exceptions:** If the device is disconnected, data sync will be retried later.

▼ 4.3 COMMUNITY ENGAGEMENT AND CHALLENGES

- **Actors:** User, Other Users, System
- **Description:** Users participate in fitness challenges and share progress.
- **Preconditions:** User must be logged in.
- **Steps:**
    1. User browses available challenges.
    2. User joins a challenge.
    3. User logs workout progress.
    4. System updates challenge leaderboard.
- **Postconditions:** Challenge participation status is updated.
- **Exceptions:** If challenge requirements are not met, the system provides reminders.

▼ 4.4 INTEGRATION WITH WEARABLE DEVICES

- **Actors:** User, Wearable Device, System
- **Description:** The app syncs data from fitness wearables.
- **Preconditions:** Wearable device must be paired with the app.
- **Steps:**
    1. User connects a wearable device.

2. System fetches real-time fitness data.
3. Data is analyzed and stored.
4. User views insights in the app.

- **Postconditions:** Data from the wearable device is available for tracking.
- **Exceptions:** If the device fails to sync, an error message is displayed.

▼ 4.5 PUSH NOTIFICATION MANAGEMENT

- **Actors:** User, System
- **Description:** Users manage notifications for reminders and goals.
- **Preconditions:** Notifications must be enabled.
- **Steps:**
  1. User configures notification preferences.
  2. System sends reminders based on schedule.
  3. User receives and interacts with notifications.
- **Postconditions:** Notifications are customized as per user preferences.
- **Exceptions:** If the app is restricted from sending notifications, the user is alerted.

▼ 5. TEST CASES

▼ 5.1 WORKOUT PLAN CUSTOMIZATION

- **Description:** Verify that users can customize and save workout plans.
- **Expected Result:** Customizations should be reflected in the user profile.

▼ 5.2 WEARABLE DEVICE INTEGRATION

- **Description:** Ensure that the app syncs data from wearables.
- **Expected Result:** Real-time data should update in the app.

▼ 5.3 SOCIAL FEATURE ENGAGEMENT

- **Description:** Verify users can participate in challenges and share progress.
- **Expected Result:** User posts and challenge progress should be displayed.

5.4 PUSH NOTIFICATION ACCURACY

- **Description:** Ensure that notifications are sent as per user settings.
- **Expected Result:** Notifications should arrive at the correct time with the right content.

5.5 DATA SECURITY COMPLIANCE

- **Description:** Validate encryption and compliance with GDPR/HIPAA.
- **Expected Result:** User data should be securely stored and transmitted.

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6. CONCLUSION

This document outlines the functional and non-functional requirements of the fitness app, ensuring a structured roadmap for developers, designers, and testers. By clearly defining the requirements, this document helps ensure that the app meets the expectations of users, stakeholders, and regulatory bodies.

The fitness app aims to provide a comprehensive and user-friendly experience by integrating advanced features such as AI-based recommendations, wearable device support, and social engagement tools. Through scalable infrastructure, security compliance, and intuitive UI/UX design, the app ensures reliability, data protection, and seamless performance.

Future enhancements may include additional AI-driven insights, deeper third-party integrations, and extended accessibility features to cater to a broader audience. As technology evolves, this document will serve as a foundation for ongoing development and improvements in the fitness tracking experience. This document outlines the functional and non-functional requirements of the fitness app, ensuring a structured roadmap for developers, designers, and testers.